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School of Computer Science and Engineering

Fall Semester 2025-26

BCSE331P – Exploratory Data Analysis

Lab Assessment 5

Multivariate Analysis

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1. **Load the dataset shared through e-mail into the R environment. Examine the dataset and identify the numerical and categorical variables.**

AIM:

To load the dataset into the R environment and identify the numerical and categorical variables.

CODE:

assessment_5.R*

Source on Save

```
#23BDS0334
#ARJUNMISHRA
data <- read.csv("assessment.csv")
head(data)
str(data)
summary(data)
sapply(data, class)
```

OUTPUT:

```
> #23BDS0334
> #ARJUNMISHRA
> data <- read.csv("assessment.csv")
> head(data)
  rownames Gunnel Time Fromlow Slope  Rw Amphiso Subst Pool Water Cobble
1        1      0  786     190    10 0.1       0     3    0     0      0
2        2      0  794     182    10 0.0       0     3    0     0      0
3        3      0  798     178    20 0.0       0     1    0     0      0
4        4      0  800     176    70 0.3       0     1    0     1      0
5        5      0  804     172    40 0.3       0     1    0     1      0
6        6      0  808     168    60 0.3       0     1    0     1      0
> str(data)
'data.frame':   1592 obs. of  11 variables:
 $ rownames: int  1 2 3 4 5 6 7 8 9 10 ...
 $ Gunnel  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Time    : int  786 794 798 800 804 808 814 817 821 826 ...
 $ Fromlow : int  190 182 178 176 172 168 162 159 155 150 ...
 $ Slope   : int  10 10 20 70 40 60 40 0 50 20 ...
 $ Rw      : num  0.1 0 0 0.3 0.3 0.3 0.3 0.9 0.5 0.1 ...
 $ Amphiso : int  0 0 0 0 0 0 0 1 0 0 ...
 $ Subst   : int  3 3 1 1 1 1 1 1 1 1 ...
 $ Pool    : int  0 0 0 0 0 0 0 1 0 0 ...
 $ Water   : int  0 0 0 1 1 1 1 1 1 0 ...
 $ Cobble  : int  0 0 0 0 0 0 0 0 0 0 ...
> summary(data)
    rownames          Gunnel             Time         Fromlow          Slope             Rw        
 Min.   :   1.0   Min.   :0.00000   Min.   :340.0   Min.   :  0.00   Min.   : 0.000   Min.   :0.0000  
 1st Qu.: 398.8   1st Qu.:0.00000   1st Qu.:623.0   1st Qu.: 71.75   1st Qu.: 0.000   1st Qu.:0.0000  
 Median : 796.5   Median :0.00000   Median :712.5   Median :158.00   Median : 0.000   Median :0.6000  
 Mean   : 796.5   Mean   :0.03015   Mean   :705.2   Mean   :149.41   Mean   : 9.504   Mean   :0.5271  
 3rd Qu.:1194.2   3rd Qu.:0.00000   3rd Qu.:810.0   3rd Qu.:224.25   3rd Qu.:10.000   3rd Qu.:1.0000  
 Max.   :1592.0   Max.   :1.00000   Max.   :983.0   Max.   :304.00   Max.   :90.000   Max.   :1.0000  
    Amphiso           Subst             Pool             Water            Cobble      
 Min.   :0.0000   Min.   : 1.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.000  
 1st Qu.:0.0000   1st Qu.: 1.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000  
 Median :0.0000   Median : 2.000   Median :0.0000   Median :1.0000   Median :0.000  
 Mean   :0.4642   Mean   : 3.106   Mean   :0.2023   Mean   :0.5088   Mean   :0.191  
 3rd Qu.:1.0000   3rd Qu.: 4.000   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:0.000  
 Max.   :4.0000   Max.   :13.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.000  
> sapply(data, class)
 rownames    Gunnel      Time   Fromlow     Slope        Rw   Amphiso     Subst      Pool     Water    Cobble 
"integer" "integer" "integer" "integer" "integer" "numeric" "integer" "integer" "integer" "integer" "integer" 
```

RESULT:

The dataset was successfully loaded and the numerical and categorical variables were identified.

2. **Create a scatter plot involving three variables, where two numerical variables are represented on the X- and Y-axes, and the third variable is represented using color.**

AIM:

To create a scatter plot using two numerical variables on the X- and Y-axes and represent a third categorical variable using color.

CODE:

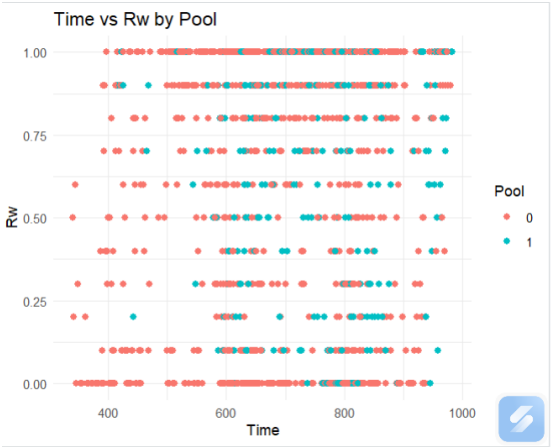
assessment_5.R*

Source on Save

```r
#23BDS0334
#ARJUNMISHRA
library(ggplot2)
data$Pool <- factor(data$Pool)
ggplot(data, aes(x=Time, y=Rw, color=Pool)) +
  geom_point(size=2) +
  labs(title="Time vs Rw by Pool",
       x="Time",
       y="Rw",
       color="Pool") +
  theme_minimal()
```

OUTPUT:

```
R 4.5.2 · ~/
> #23BDS0334
> #ARJUNMISHRA
> library(ggplot2)
> data$Pool <- factor(data$Pool)
> ggplot(data, aes(x=Time, y=Rw, color=Pool)) +
+   geom_point(size=2) +
+   labs(title="Time vs Rw by Pool",
+        x="Time",
+        y="Rw",
+        color="Pool") +
+   theme_minimal()
>
```



RESULT:

The relationship between Time and Rw was successfully visualized according to different Pool categories.

3. **Create a scatter plot involving four variables, where two numerical variables are represented on the X- and Y-axes, the third variable is represented using color, and the fourth variable is represented using shape.**

AIM:

To create a scatter plot using two numerical variables, one categorical variable represented by color, and another categorical variable represented by shape.

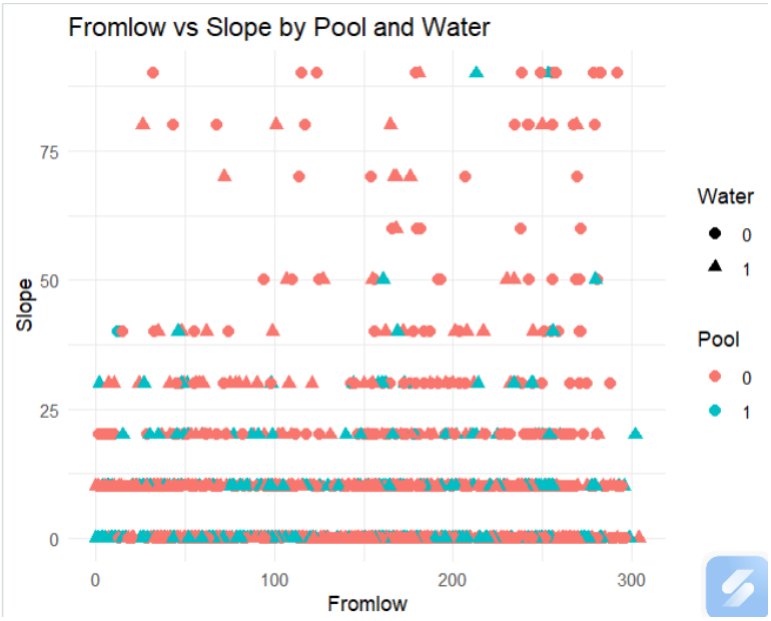
CODE:

assessment_5.R* ×

Source on Save

```
#23BDS0334
#ARJUNMISHRA
data$Pool <- factor(data$Pool)
data$Water <- factor(data$Water)
ggplot(data, aes(x=Fromlow, y=Slope,
                 color=Pool,
                 shape=Water)) +
  geom_point(size=2.5) +
  labs(title="Fromlow vs Slope by Pool and Water",
       x="Fromlow",
       y="Slope",
       color="Pool",
       shape="Water") +
  theme_minimal()
```

OUTPUT:



RESULT:

The relationship between Fromlow and Slope was successfully visualized based on Pool and Water categories.

4. **Create a scatter plot involving four variables, where two numerical variables are represented on the X- and Y-axes, the third variable is represented using color, and the fourth numerical variable is represented using point size.**

AIM:

To create a scatter plot where two numerical variables are represented on the X- and Y-axes, a categorical variable is represented using color, and a fourth numerical variable is represented using point size.

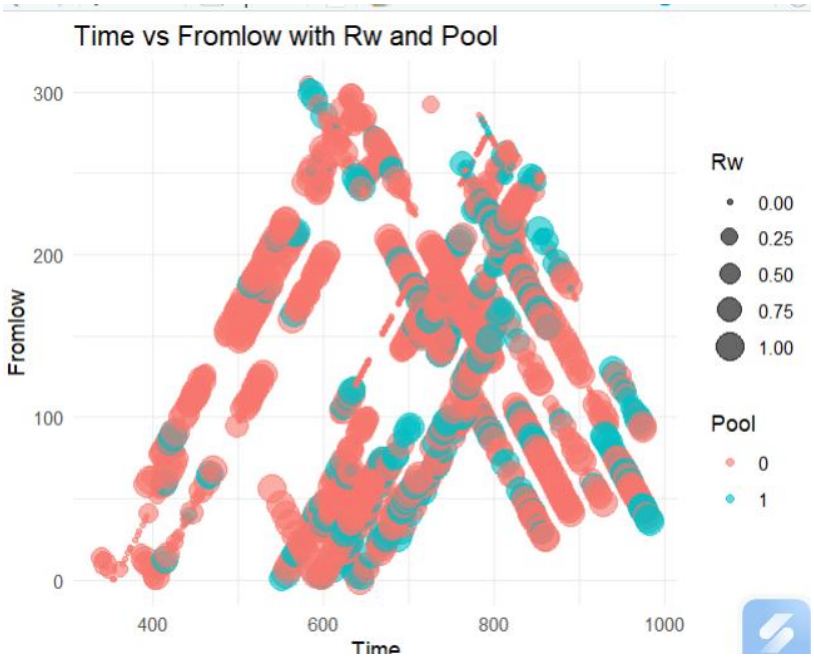
CODE:

assessment_5.R*

Source on Save

```
#23BDS0334
#ARJUNMISHRA
data$Pool <- factor(data$Pool)
ggplot(data, aes(x=Time, y=Fromlow,
                 color=Pool,
                 size=Rw)) +
  geom_point(alpha=0.6) +
  labs(title="Time vs Fromlow with Rw and Pool",
       x="Time",
       y="Fromlow",
       color="Pool",
       size="Rw") +
  theme_minimal()
```

OUTPUT:



RESULT:

The relationship among Time, Fromlow, Rw, and Pool was successfully visualized using color and point size.

5. **Construct a bubble plot using three numerical variables, where two variables are represented on the X- and Y-axes, and the third variable determines the size of the bubbles.**

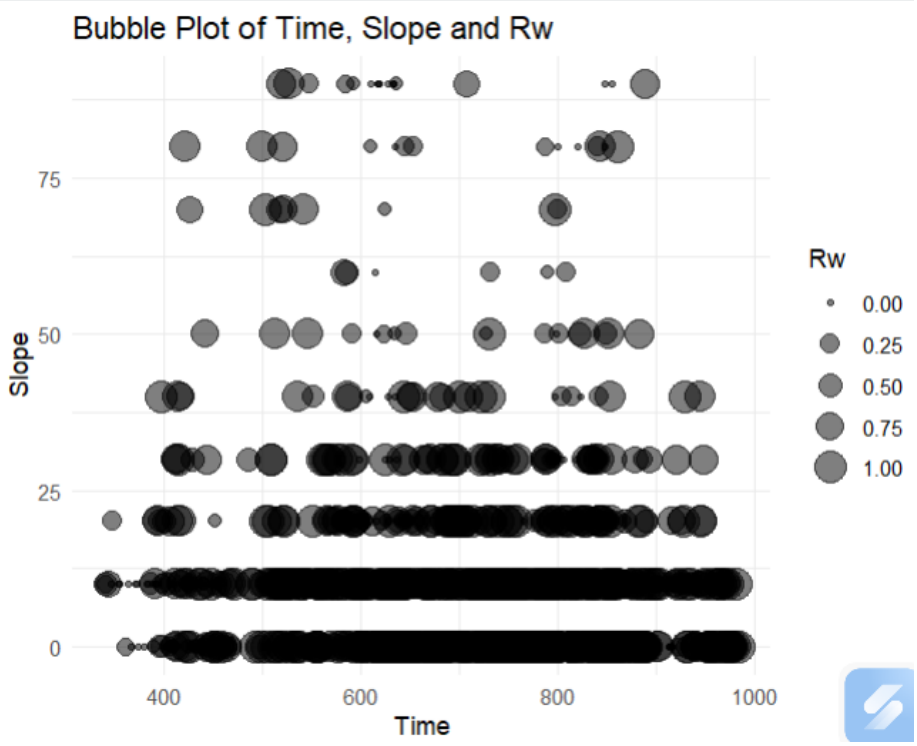
AIM:

To construct a bubble plot using three numerical variables, where two variables are represented on the X- and Y-axes and the third determines bubble size.

CODE:

```
#23BDS0334
#ARJUNMISHRA
ggplot(data, aes(x=Time, y=Slope, size=Rw)) +
  geom_point(alpha=0.5) +
  labs(title="Bubble Plot of Time, Slope and Rw",
       x="Time",
       y="Slope",
       size="Rw") +
  theme_minimal()
```

OUTPUT:



RESULT:

The three numerical variables were successfully represented using a bubble plot.

6. **Generate subplots using facet_wrap() by selecting an appropriate categorical variable from the dataset.**

AIM:

To generate separate subplots using facet_wrap() based on an appropriate categorical variable.

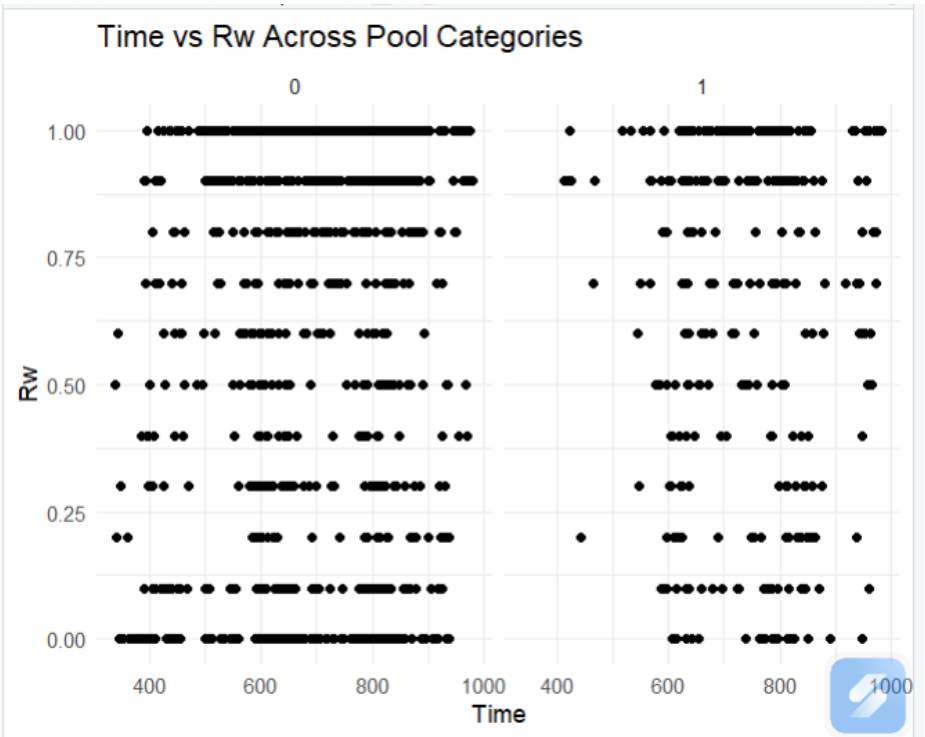
CODE:

assessment_5.R*

Source on Save

```
#23BDS0334
#ARJUNMISHRA
data$Pool <- factor(data$Pool)
ggplot(data, aes(x=Time, y=Rw)) +
  geom_point() +
  facet_wrap(~Pool) +
  labs(title="Time vs Rw Across Pool Categories",
       x="Time",
       y="Rw") +
  theme_minimal()
```

OUTPUT:



RESULT:

The relationship between Time and Rw across different Pool categories was successfully compared using facet_wrap().

7. **Generate subplots using facet_grid() by selecting two appropriate categorical variables from the dataset. Arrange the plots in rows and columns and compare the patterns observed across the different groups.**

AIM:

To generate subplots using two categorical variables and arrange them into rows and columns using facet_grid().

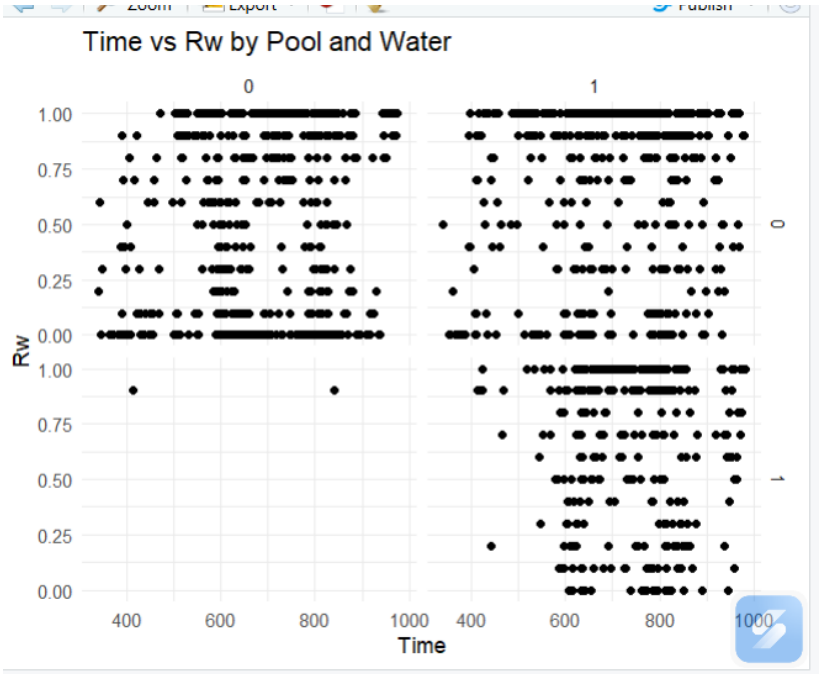
CODE:

assessment_5.R*

Source on Save

```
#23BDS0334
#ARJUNMISHRA
data$Pool <- factor(data$Pool)
data$Water <- factor(data$Water)
ggplot(data, aes(x=Time, y=Rw)) +
  geom_point() +
  facet_grid(Pool ~ Water) +
  labs(title="Time vs Rw by Pool and Water",
       x="Time",
       y="Rw") +
  theme_minimal()
```

OUTPUT:



RESULT:

The patterns between Time and Rw were successfully compared across combinations of Pool and Water using facet_grid().

8. **Construct a pairs plot using at least four numerical variables from the dataset. Use the plot to identify possible relationships and correlations among the selected variables.**

AIM:

To construct a pairs plot using at least four numerical variables and identify possible relationships and correlations among them.

CODE:

assessment_5.R* | num_data

Source on Save

```
#23BDS0334
#ARJUNMISHRA
num_data <- data[,c("Time","Fromlow","Slope","Rw")]
pairs(num_data,
      main="Pairs Plot of Numerical Variables",
      pch=19)
```

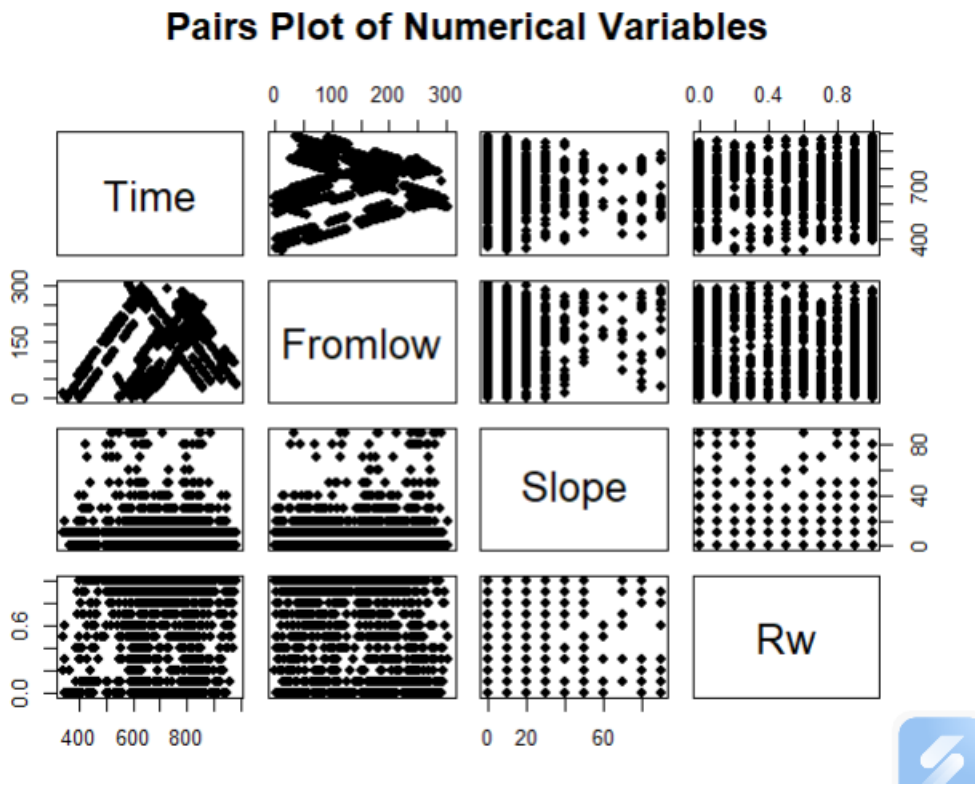
OUTPUT:

assessment_5.R* | num_data

Filter

	Time	Fromlow	Slope	Rw
1	786	190	10	0.1
2	794	182	10	0.0
3	798	178	20	0.0
4	800	176	70	0.3
5	804	172	40	0.3
6	808	168	60	0.3
7	814	162	40	0.3
8	817	159	0	0.9
9	821	155	50	0.5
10	826	150	20	0.1
11	829	147	0	0.7
12	834	142	10	0.9
13	839	137	0	0.3
14	842	134	0	0.0
15	844	132	10	0.0
16	845	131	20	0.9
17	849	127	50	0.4
18	851	125	50	0.9
19	854	122	0	0.8
20	855	121	30	0.9

Showing 1 to 21 of 1,502 entries, 4 total columns



RESULT:

The pairs plot was successfully generated to examine possible relationships and correlations among the numerical variables.

9. **Create a parallel coordinate plot using at least four numerical variables from the dataset. Differentiate the observations using an appropriate categorical variable.**

AIM:

To create a parallel coordinate plot using at least four numerical variables and differentiate observations using a categorical variable.

CODE:

assessment_5.R*

Source on Save

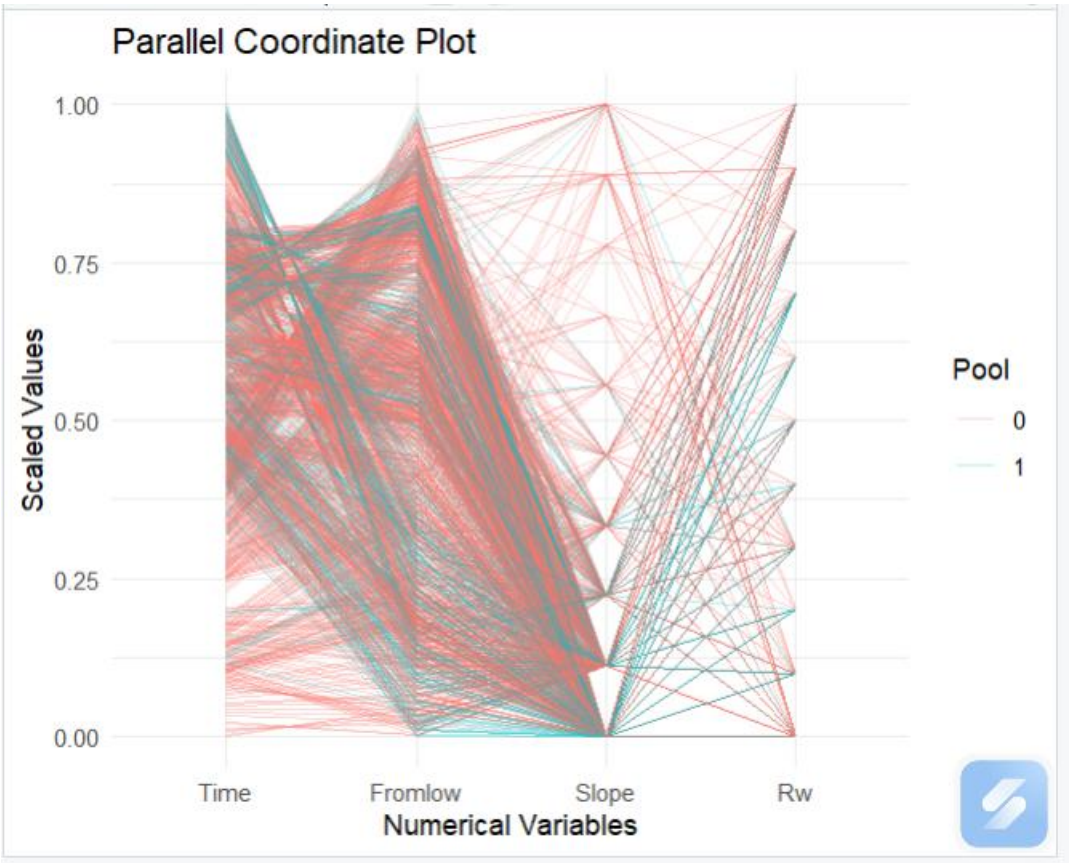
```
#23BDS0334
#ARJUNMISHRA
library(ggplot2)
library(GGally)
data <- read.csv("assessment.csv")
data$Pool <- factor(data$Pool)
pdata <- data[, c("Time", "Fromlow", "Slope", "Rw", "Pool")]
ggparcoord(pdata,
           columns=1:4,
           groupColumn=5,
           scale="uniminmax",
           alphaLines=0.3) +
  labs(title="Parallel Coordinate Plot",
       x="Numerical Variables",
       y="Scaled Values",
       color="Pool") +
  theme_minimal()
```

OUTPUT:

assessment_5.R* | num_data

Filter

	Time	Fromlow	Slope	Rw
1	786	190	10	0.1
2	794	182	10	0.0
3	798	178	20	0.0
4	800	176	70	0.3
5	804	172	40	0.3
6	808	168	60	0.3
7	814	162	40	0.3
8	817	159	0	0.9
9	821	155	50	0.5
10	826	150	20	0.1
11	829	147	0	0.7
12	834	142	10	0.9
13	839	137	0	0.3
14	842	134	0	0.0
15	844	132	10	0.0
16	845	131	20	0.9
17	849	127	50	0.4
18	851	125	50	0.9
19	854	122	0	0.8
20	855	121	30	0.9



RESULT:

The parallel coordinate plot was successfully created to compare patterns across the selected numerical variables and Pool categories.